

Platformisation of Digital Financial Services (DFS): The Journey of DFS in the Global North and Global South

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ABSTRACT

The digitization of financial transactions in both Global North and Global South has led to considerable shifts in how money is used, understood, and processed by users, banks, and fintechs. This shift from physical cash to digital media, accelerated by the COVID-19 push for digital transactions, has impacted how users perceive and use digital money and opened avenues for more data collection. This diverse panel proposes a discussion to understand the set of opportunities and challenges around the design of digital financial services (DFS) and data-driven decision-making in DFS. We will create a live working document starting before the panel to document the discussion, which develops during and after the panel. This live document will enable community to engage with a broader audience of researchers and industry, outlining processes, methods,

and tools that researchers and practitioners have created to work with users to develop new equitable DFS and further exploration.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing.**

KEYWORDS

Digital Financial Services, Financial Inclusion, Algorithmic Decisions, Fintech

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1 INTRODUCTION

Providing access to the necessary and affordable financial products and services and addressing financial exclusion has been a

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prime objective for most countries around the world [12, 15, 30]. According to the World Bank, people ought to be able to access the financial services and products that are suited for them and fulfil their individual requirements, regardless of their background or income [5]. Being financially included is critical throughout a person's financial life. Individuals and businesses can utilise a range of acceptable financial services, such as savings, payments, credit, and insurance, with financial access as the first step toward greater financial inclusion [11].

In the 2030 Sustainable Development Goals, financial inclusion is highlighted as a critical facilitator of other development goals, with targets in eight of the seventeen goals. SDG 1 is about ending poverty; SDG 2 is about ending hunger, achieving food security, and promoting sustainable agriculture; SDG 3 is about promoting health and well-being; SDG 5 is about gender equality and women's economic empowerment; SDG 8 is about promoting economic growth and jobs; SDG 9 is about supporting industry, innovation, and infrastructure; and SDG 10 is about reducing inequality. Furthermore, there is an implied role for increased financial inclusion in SDG 17 on improving the means of implementation, which may boost growth by increasing savings mobilisation for investment and consumption [3, 20].

According to the Global Findex Database, while 1.2 billion individuals have created a financial account since 2011, an estimated 1.7 billion adults (or 31% of adults) still do not have a basic transaction account [11]. More than 60% of employees are employed by small and medium-sized organisations (SMEs), which account for 95% of all businesses globally. However, SMEs also face financial challenges. Fintech (financial technology innovation) is offering enormous potential in this context, aided by the increased ownership of mobile phones and internet connectivity [27]. By lowering or eliminating the need for physical encounters and the need for cash, fintech is enabling governments reach—quickly and securely—people and companies with different types of income and liquidity support.

The financial sector is being transformed by technology, which is giving unprecedented access to financial services. These changes have been in the works for a number of years and are affecting nearly every country on the world. Despite social distance and containment attempts, technology has opened up new possibilities for digital financial services to speed up and increase financial inclusion during the COVID-19 outbreak. Simultaneously, as digital financial services develop, the threats that appeared before to COVID-19 are growing more serious. These threats could be broadly divided into two parts:

- **First**, there is a constant tension between what is available in terms of digital payment artefact design and what would constitute *appropriate design* for users who are disadvantaged in terms of income, education, and technology exposure (also known as Emergent Users [4]), who may adopt digital payments and other services.
- **Second**, data has long been utilised by financial services firms to aid in decision-making, such as lending and investing. Data biases, as well as inaccurate and insufficient information, can exacerbate financial exclusion and encourage

distrust in decision-making along with the use of new technologies. A lack of financial and digital understanding may heighten these risks.

2 PANEL OBJECTIVES

Banks and their agents, microfinance institutions, and informal networks (for example, relying on family, micro-lending clubs, or moneylenders) have traditionally provided financial services with little competition [25]. They are primarily based on cash transactions and in-person encounters with financial service providers. These contacts serve as the foundation for creditworthiness monitoring and are frequently used to teach clients about money. The introduction of fintech is transforming this landscape: the development of digital financial instruments available through mobile phones or computers has considerably decreased the necessity for face-to-face encounters. The current COVID-19 pandemic's movement constraints have increased the benefits of increasing digital banking services [22]. The creation of digital platforms that can offer a variety of financial products and act as aggregators for existing financial products or fintech companies' own products helps customers get the most out of their money by comparing the price and suitability of products and services offered by different companies [1].

Since long before the COVID-19 crisis, global leaders and policymakers have been aware of fintech's potential to increase financial inclusion. The Alliance for Financial Inclusion (AFI), a global network of policymakers, was founded in 2008 and released the Maya Declaration in 2011 [10]. At the G20 Summit in Seoul in 2010, the leaders emphasised financial inclusion by supporting a Financial Inclusion Action Plan (FIAP) [29] and establishing the Global Partnership for Financial Inclusion (GPFI) [14]. The AFI and GPFI recognised technology as a critical component of financial inclusion in 2016, launching a new work stream called Fintech for Financial Inclusion [2]. The COVID-19 pandemic has cast a light on how digital financial inclusion can be used to respond to crises and how crises may drive digital financial inclusion. Hence, the panel would discuss two primary objectives to respond.

- **From design perspective:** When it comes to transitioning unbanked segments from informal analogue contacts to formal internet platforms, what makes an *appropriate design* will be challenging. COVID-19 restrictions, technology usage and penetration in users' lives, users' life priorities (food on the table vs individual devices), the need for technological inclusiveness, and digital literacy are among few of the challenges.
- **From data perspective:** Massive data collection, breakthroughs in computer algorithms, and gains in processing power provided by high-speed internet and artificial intelligence in digital financial services may present both opportunities and challenges. New potential difficulties include unequal access to digital infrastructure, financial and digital literacy limitations, and possible biases reinforced by new data sources and analytics.

3 PANEL OUTCOMES

The field of financial HCI is new, but there is a growing concern to tackle this challenge in the CHI community. This panel will bring together researchers and practitioners from both academia and industry who are working in the financial HCI. Through panel, we aim to bring together different people working on financial HCI to share their work and coherently formulate ways to overcome the challenges while designing for low-income individuals.

The purpose of this panel is to engage researchers, industry, and the broader CHI community in the discussion and debate around digital financial services in the COVID era. In doing this, we aim to address a number of key aims by exploring:

- How emerging payments options — across different cultural, policy, and infrastructural dimensions — during the COVID-19 crisis have enabled people to pay without cash? How has this affected people's behaviour regarding contactless payment; what are the interactional, transactional, social, and other challenges in contactless payment?
- How digital business models that also use data generated by digital transactions and enriched digital financial data, have led to the creation of novel financial products and services, and the integration of traditional financial infrastructures (such as banks) with technological-driven companies such as Paytm and Apple Pay, leading to new forms of platformisation?

3.1 Research Questions

We will start by exploring the current challenges and opportunities in financial HCI community. For example:

- Where was digital financial inclusion emerging prior to the COVID-19 crisis? Is fintech increasing financial inclusion?
- Are fintech companies disrupting traditional providers, and how could these relationships evolve during the ongoing downturn and the subsequent recovery?
- How can digital artefacts for financial services and payments be designed so that they reflect the end user's socio-financial circumstances? How has it impacted on the economic and social lives of women?
- How to design digital payments for fragmented infrastructures, heterogeneous devices, intermittent connectivity, functional and financial illiteracy?
- More data and better algorithms, in principle, might improve the accuracy of these forecasts. Simultaneously, as stakeholders and consumers grow less able to grasp how an algorithm arrives at its output and what assumptions it makes about an individual in doing so, sophisticated algorithms may raise the risk of indirect bias via proxy. What are some options for resolving this problem?

4 PANELISTS

We bring diverse panelists from academia and industry-based out of many regions worldwide in this panel. The panelists have vast experience working in their contexts and have knowledge and awareness of other contexts. There is the representation of both the Global North and the Global South. Hence, differing opinions are expected and appreciated. Therefore, this panel proposal is not

necessarily reflective of panelist opinions, and due to the intention for discussion and intellectual provocation on the mentioned topics, there is no expectation that panelists will reach the unilateral consensus of any form.

4.1 Moderators

- **Alex Blandin (Academia)** is a computer science researcher and doctoral student at Swansea University, with a current focus on Explainable Artificial Intelligence (XAI) and Fair Machine Learning (FairML). This is being explored within the context of financial and banking systems that may benefit from AI and ML, such as for decision support, wherein the aim is to produce human-centred systems that are approachable and trustworthy. Due to the increasing appeal of data-driven systems and decision-making to deliver new capabilities and conveniences, the goal of such research is to ensure that these systems do not leave the human element behind and preserve the trust typically invested in comparable human processes.
- **Matt Jones (Academia)** is the founding Director of the Morgan Advanced Studies Institute (MASI), Wales' first institute of its kind (www.swansea.ac.uk/masi). His HCI work has focused on emerging technologies (such as speech assistants, haptics, computational materials) and emergent contexts (particularly in sub-Saharan Africa and India). He helped to build the Mobile HCI community and wrote two books and many articles on the subject (more at www.undofuture.com). He was dazzled by the luminaries, and overwhelmed by the food, at the HCI 2020 forum: he still sizzles due to the experience.
- **Matt Roach (Academia)** is a Senior Lecturer in Computer Science at Swansea University. Matt is an investigator on projects applying machine learning to the challenges of smart city traffic management and fraud detection. Matt has extensive experience in collaborative research and leading business engagement activities on Trust, Identity, Privacy and Security in the domains of Health & Wellbeing, Smart Cities, and Digital Economy. His research interests are multi-disciplinary and focus on the intersection of people and data, themes include: Machine Learning; Algorithmic Fairness; Pervasive & Persuasive Computing; Human Computer/Data Interaction; Participatory Design; Socio-Technical Decision-Making Systems; Socio-Technical Behaviour Change Systems.
- **Pranjal Jain (Academia)** is a post-graduate research student at Swansea University, UK. His work involves researching on data privacy in every day and digital financial services [18, 19]. He is trained in Human-centred Design. His research interests are broadly in Human-Computer Interaction (HCI) and User Experience (UX). He holds a Master in Design (MDes) in Human-centred Design from Srishti Institute of Art, Design and Technology, and Bachelor in Technology (BTech) in Electronics and Communications from the Vellore Institute of Technology, Vellore.

4.2 Invited Speakers

- **Mark Perry (Academia)** is an interdisciplinary user studies researcher. His work involves exploring how digital technology is practically used by people in support of interaction design, with research covering the areas of mobile and ubiquitous technology, digital and mobile money, and emerging financial services (FinTech). He has worked on a number of research projects around digital connectivity and peer-to-peer relationships in financial services, exploring user experience, stakeholder perspectives, community development, and systems design. Most recently, he has been exploring the social impacts of payments [21, 26] and financial technologies in China.
- **Samia Ibatasam (Academia)** is a Ph.D. student in the Paul G. Allen School of Computer Science & Engineering at the University of Washington, Seattle. Her research explores the intersection of gender in women's financial and technological inclusion [16, 17]. As part of the DFS research group funded by BMGF, her work in Pakistan has explored the learnability of existing mobile wallets, women's financial knowledge and decision-making, and SMS based financial frauds. Before UW, she was a faculty member at the Information Technology University in Lahore, Pakistan and was also the founding Co-Director of Innovations for Poverty Alleviation Lab (IPAL). She worked to design solutions for Maternal, Neonatal and Child Health (MNCH).
- **Suleman Shahid (Academia)** is an Assistant Professor in Computer Science at the Lahore University of Management Sciences (LUMS), Lahore, Pakistan where he directs the Computer Human Interaction and Social Experience Lab (CHISEL). His research focuses at the intersection of design, technology and wellbeing with applications to assistive technologies to enhance the quality of life of persons with disabilities and persons with mental health conditions, child-computer interaction, and human computer interaction for development (HCI4D).
- **David Sullivan (Industry)** is a Lead Data Scientist at Starling Bank. He earned his PhD from the University of Sussex in Astronomy and Astrophysics, where he first made use of machine learning to improve galaxy formation models, and has continued to explore applications of machine learning ever since. His work at Starling has helped them create new schemes that raise financial inclusivity, such as their Connected Cards and in-app cheque deposits, or promote algorithmic fairness and exploration of how to further improve fairness in their data-driven systems and decision-making.
- **Jacki O'Neil (Industry)** is Microsoft Africa Research Institute (MARI) Director. She is passionate about designing technologies which enhance, rather than remove, agency and create sustainable futures. She brings this passion to the MARI where she is building a multidisciplinary team, combining research, engineering and design to solve local problems globally. An ethnographer by trade, in her research career so far she has focused on technologies for work – with the aim of making work better; and technologies for societal impact, with the aim of supporting underserved communities [13, 23, 24, 28].
- **James Pomeroy (Industry)** is a global economist at HSBC. He joined the Economics team in 2013 having previously worked within the Asset Allocation research team. His global work focuses on longer-term trends and themes, and the impact that they have on the economy and policy decisions today. Demographic data is at the heart of much of his work, but he has also written about urbanisation, the role of technology in the economy and how the world is moving away from cash. Alongside this, he provides economics coverage of Scandinavia. James holds a BSc in Economics from the University of Bath.
- **Heloisa Candello (Industry)** is a research scientist in the Tech4Impact group of IBM Research laboratory in Brazil. Her work focuses on human and social aspects of Artificial Intelligence systems, particularly conversational user interfaces. Her work in this area includes the development and application of mixed-methods research in the collection, design, and evaluation of conversational systems. Currently, Heloisa is leading a project that aims to bring “conscious” access to microcredit by enhancing non-traditional financial practices of low-income small business owners with AI technology in the Global South [6–8].
- **Rajat Jain (Industry)** is a seasoned product and business leader with domain expertise in fintech and payments. Currently, he's leading the product initiative on building monetisation and lending at BharatPe. Previously, Rajat has worked on Digital India initiatives enabling businesses digitise customer onboarding and servicing. He was heading product management at Khosla Labs, a Khosla Ventures backed company. Prior to Khosla Labs, Rajat helped set up the Financial Services vertical at Paytm where he built consumer lending products from conceptualisation to execution. He built Paytm Postpaid, Seller Financing and Salt from the ground up.
- **Robert Dowd (Industry)** is a Yimba's Founder and Chief Executive Officer. He has worked in the financial services market for over 20 years. He has built a sophisticated but simple solution to engage consumers in the use of Digital Wallets. In 1999, built one of the first digital marketing CRM systems for Marketing Agencies. Delivered via a cloud based platform it eliminated the need for the Agencies to build the capability in-house. Successfully sold the company in 2004 having contracted with four agencies and numerous institutional clients.
- **Beni Chigh (Non-profit)** manages the research at the Future of Finance Initiative. Her work focuses on identifying systemic stability and consumer protection concerns in digital finance [9]. She also contributes to the Social Protection Open Digital Ecosystem work stream at Dvara Research, where her work focuses on designing and implementing citizen centric, digital social protection delivery platforms. Beni holds a Masters in Public Policy from NLSIU, Bangalore and an M.A in Development Economics from the South Asian University, New Delhi.

5 PANEL FORMAT AND AUDIENCE ENGAGEMENT

The panel will begin with brief (3-4 minute) position statements from each panelist, focusing on what they see as the key challenges and opportunities for DFS and research. To ground the conversation, panelists will show examples of today's financial transactions systems and discuss design choices made concerning how these financial systems interact with humans. There will also be a live Q&A session in which audience members can pose questions to the entire panel. Before, during and after the event, the audience members and panelists will participate in an online group discussion facilitated by the moderators on Miro Board and Word Document. We will invite the audience to use this discussion forum to inspire the community with their thoughts on the future of Digital Financial Services. We plan to host an after panel party for virtual and in-person participants to steer the conversation informally. We plan to report on the discussion in the panel session and in the online discussion to Interactions, Communications of the ACM, a special issue of a journal, or possibly a CHI 2023 submission. We would use the following web link to engage with the broader audience: <https://cs-web.swansea.ac.uk/ai-in-finance/>.

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